

Sahil Khan

✉ s.sahil9752@gmail.com | 📞 +91 9752588937 | 🌐 github.com/sahell0x
🌐 linkedin.com/in/sahell0x | 📄 leetcode.com/Sahell | 🌐 portfolio.sahellx.site

Skills

Languages: JavaScript, Python, C/C++

Technologies & Tools: Express, FastAPI, AWS (EC2, S3, Lambda), Docker, vLLM, LangChain, Postgres, Pinecone, Redis, LangGraph, RAG, Node.js, MongoDB, WebSocket, Git-GitHub, React.js, Tailwind

Work Experience

Purple Sky Infotech

Jun 2025 - Present

Software Engineer

- Architected a high-throughput, “true natural” voice agent platform using Pipecat, orchestrating diverse STT/TTS engines and LLM APIs (OpenAI, Gemini) to deliver conversational AI with sub-second latency.
- Developed a robust backend service utilizing FastAPI to manage multi-tenant agent sessions as a service, integrating telephony providers like Twilio and Plivo to scale concurrent active calls by over 60%.
- Deployed and optimized open-source LLMs on RunPod GPU servers (RTX 3090) using vLLM, achieving a 40% reduction in inference costs while maintaining high-availability real-time latencies.
- Built a high-performance Retrieval-Augmented Generation (RAG) system using LangChain, Pinecone, and OpenAI’s text-embedding-3-small model, enabling contextual agent responses over 100,000+ document chunks with sub-200ms query latency.
- Python, FastAPI, vLLM, RunPod, Pipecat, Twilio, Plivo, OpenAI, Gemini, Docker, LangChain, Pinecone

Open Source - Sparrow (Techdome)

Core Contributor

- Added API response download functionality and developed seamless image response handling/UI rendering for REST APIs, ensuring 100% accurate processing of binary image data.
- Implemented WebSocket disconnection detection with real-time UI alerts, ensuring 100% accuracy in identifying dropped connections and improving platform stability.
- TypeScript, React.js, WebSocket, REST APIs, Git-GitHub

Affimintus Technologies, Indore

Jun 2024 - Nov 2024

Software Developer Intern

- Reduced initial page load time by 25% and integrated REST APIs efficiently, reducing average response time by 20% for key user flows.
- Collaborated with the backend team to implement JWT-based authentication in Express and used Recoil for state management, improving app scalability and significantly reducing state-related bugs.
- Node.js, Express.js, JWT, Recoil, React.js, HTML, CSS, Agile

Project Work

- **FeelChat - Secure real-time emotion-based Chat app:** Developed an end-to-end encrypted chat application that revolutionizes conversations with real-time emotion sharing. Integrated a lightweight machine learning model directly in the browser via face-api.js to detect facial expressions, ensuring absolute user privacy. Scaled application communications using Socket.io and managed complex UI states efficiently with Recoil, achieving sub-100ms processing times and zero-lag synchronization for concurrent users. React.js, TypeScript, Recoil, Express.js, face-api.js, Socket.io, MongoDB.
- **Code Atopsy - AI-powered Code Analysis and Visualizer Tool:** Engineered a web-based artificial intelligence assistant designed to visually break down and audit algorithms across multiple programming languages. Utilized custom AI agents and advanced prompt engineering to generate deep contextual insights, architectural explanations, and active debugging suggestions. Improved code comprehension for developers, effectively reducing manual debugging and code review time by over 30%. React.js, Express.js, TypeScript, Node.js, Tailwind, Custom AI agent.

Education

Jawaharlal Institute of Technology Vidhya Vihar Borawan
B.E. in Electronics and Communication Engineering

Graduated 2025
CGPA: 7.34/10